Sustainable Grid Stabilization via Green Hydrogen Technologies

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***Abstract*— Renewable energy sources are rapidly reshaping the U.S. electricity grid, but their intermittent nature leads to supply and demand mismatches, causing grid instability and economic inefficiencies. While batteries provide short-term energy storage, large-scale, long-duration solutions like hydrogen storage are essential to managing seasonal fluctuations effectively. Multiple pathways were evaluated, including AEM, PEM, and solid oxide fuel cells, as well as synthetic fuels like ammonia. Additionally, various hydrogen storage options were considered, such as abandoned oil wells, salt caverns, cryogenic tanks, and pressurized gas storage systems. A multi-step system was determined to be the most effective solution, utilizing a PEM electrolyzer to convert excess energy into hydrogen gas, storing it in a pressurized tank, and reconverting it to electricity with a PEM fuel cell during peak demand. To model the impact of the large-scale solution, an algorithm was developed to assess both residential (El Paso) and utility-scale solar operators (Imperial Irrigation District). The program determined that the optimal quantity of hydrogen gas sold to the market is 70%, utilizing forty 1500 kg storage tanks for the large scale, and ten for the small scale. The algorithm simulation determined that IID saves 14.25 million MWh of energy a year, and El Paso saves 2.39 MWh annually after storing excess nergy as hydrogen and using it in high demand periods.**

I. Introduction

In recent years, renewable energy (RE) sources have begun to play an increasingly dominant role in the generation of electricity in the United States, signaling a transformative shift in the country's energy landscape. Historically, the U.S. relied heavily on fossil fuels like coal, natural gas, and oil for power. However, more than half of the electric-generating capacity in 2023 will be solar [1], and the International Energy Agency (IEA) predicted that renewable energies will meet more than 90% of the increase in global demand by 2025 [2]. A key challenge with RE, despite its growing popularity, is that production often doesn't align consistently with energy demand, causing imbalances that affect grid efficiency and economic stability. Renewable sources, such as wind and solar, generate electricity based on environmental conditions rather than demand fluctuations, meaning peak production can occur during low-demand periods, such as during midday sunlight or nighttime winds when consumption is lower. And vice versa, having high demand at low production times like nighttime. To manage surplus energy, battery and green hydrogen technologies are being considered [3].

Batteries (BS) and hydrogen storage (GHS) each have distinct advantages and tradeoffs in energy storage . Batteries are highly efficient (~90%) and excel at providing quick bursts of power, making them ideal for short-term energy needs [3]. However, they are expensive for large-scale, long-duration storage. In contrast, hydrogen storage is less efficient (~45%) but much cheaper for storing vast amounts of energy over long periods, making them ideal for addressing seasonal mismatches between energy supply and demand [3].

California is projected to need approximately 52,000 MW of storage by 2045 to meet its energy goals [4]. Given the

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scale of the storage needed, hydrogen technologies offer a promising solution to complement batteries. The system integrates an electrolyzer for hydrogen production and a fuel cell for power regeneration, supported by a pressurized tank as hydrogen storage system. During periods of excess energy that would otherwise be curtailed, the surplus electricity powers the electrolyzer, where water undergoes electrolysis, producing hydrogen (H_2) and oxygen (O_2) gases. The hydrogen and oxygen are then stored in pressurized tanks. When grid demand rises, the stored hydrogen is fed into the fuel cell, where it generates electricity, as illustrated in Figure 1.

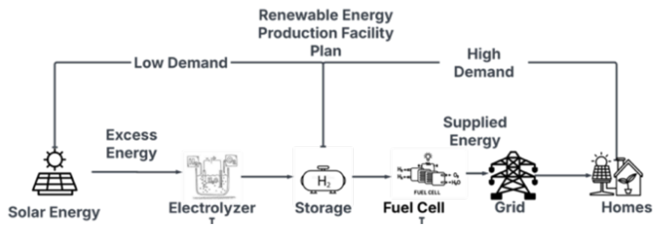


Fig. 1. Flow Diagram for Energy Storage and Energy Production using Green Hydrogen.

This article explores the potential of hydrogen-based solutions to stabilize renewable energy output, transforming intermittent sources into firm, dispatchable power. By addressing the issue of energy variability, these innovations contribute to greater grid reliability and also support broader efforts to reduce carbon emissions and improve energy sustainability.

II. Technologies Evaluation

To incorporate hydrogen-based technologies into an energy storage system, hydrogen fuel cells, electrolyzers and hydrogen storage methods were evaluated to determine their suitability, efficiency, and scalability for integration into energy systems.

A. Electrolyzer/Hydrogen Fuel Cells

There are four different main types of electrolyzers/fuel cells. Alkaline electrolyzers use concentrated alkaline solution, such as potassium hydroxide. The solution moves hydroxide ions towards the anode, producing hydrogen gas at the cathode. Alkaline electrolyzers have been a well-established technology in the industry, and because it does not require precious metals resistant to a lot of corruption making it more cost efficient. These electrolyzers also have a long lifetime of about 90,000 hours [1]. Alkaline electrolyzers operate at low currents and low temperatures (30-80˚C). [1]. These conditions limit the current densities an alkaline electrolyzer can take, which cause the electrolyzer to have a lower hydrogen production output. Alkaline electrolyzers have a slow start up, between 20 to 60 minutes, and a response time in the order of seconds, which impacts its energy efficiency and load variability [2]. Alkaline electrolyzers have an energy efficiency between 59-70%. The efficiency increases for larger hydrogen loads, making it a competitive option in large scale hydrogen production. Alkaline electrolysis produces low hydrogen purity [1] since oxygen can dissolve through the diaphragm and contaminate the hydrogen gas. Currently, further development to tackle these issues are still in the research phase but can make alkaline electrolyzers reach similar efficiencies to other electrolysis technologies.

Similar to alkaline electrolyzers are anion exchange membrane electrolyzers (AEM). These electrolyzers still use a basic solution, but instead of a diaphragm for separation, an anion exchange membrane separates the nodes. The membrane increases the hydrogen purity by rejecting the cations and only allowing anions to flow through. This design comes with its own advantages, namely the use of low concentration electrolytes compared to high concentration electrolytes, and the use of cost effective metals for the anodes and cathodes [1]. However, AEM electrolyzers are still in the research phase, and studies on the stability of AEM at large scales are still needed. Proton exchange membrane electrolyzers (PEM) use a proton exchange membrane, moving hydrogen ions from the anode to the cathode through the membrane to create hydrogen gas. PEM electrolysis operates at similar temperatures as alkaline electrolysis, but at high currents. PEMs produce high purity hydrogen gas and have a quick start up time [3].In comparison to alkaline electrolyzers, PEMs take 5 minutes to start up and their response time is in the order of milliseconds. This increases the efficiency, making PEM electrolyzers to have an efficiency of 65 to 82 percent [1]. The quick start up time also makes PEM electrolyzers more compatible with renewable energy sources. Because PEM electrolyzers use acidic solutions, they require metals that are resistant to corrosion such as platinum which makes them expensive. The corrosive liquids reduce the overall lifetime to 60,000 hours [1] compared to alkaline electrolyzers with 90,000 hours. However, because of its high current and hydrogen production output, PEM electrolyzers have been produced at large scales for industrial use.

Another option would be solid oxide electrolyzers that operate by using steam for water electrolysis. By using steam, the operating temperatures required need to be high (600-900˚C) [4], and unless steam is already a byproduct of a different system, this process may become very energy intensive. Because the water molecules are in a gaseous form, the power needed for splitting the water molecules reduces, allowing solid oxide electrolysis to have the highest efficiency at 90% [1]. Solid oxide electrolyzers do not use precious metals, making the initial costs cheaper than PEMs. However, solid oxide electrolyzers have long term stability issues that are preventing its usage industrially. With a lifetime of only 20,000 hours, it has the lowest lifetime out of all the electrolysis technologies. Because of this issue, solid oxide electrolysis is still under development.

B. Hydrogen Storage

The options considered for hydrogen gas storage include pressurized tanks, repurposing abandoned oil wells to store hydrogen, or liquifying hydrogen. One method is the use of pressurized tanks. Pressurized tanks can maintain the purity of the hydrogen gas, which would improve the efficiency of the electrolyzers. Pressurized tanks can also be transported and scaled easily, as the tanks can vary by sizes with different capacities and can be transported easily. The downsides for pressurized tanks include high initial costs and limited capacity. Because hydrogen gas requires high pressure to store, pressurized tanks can often be expensive and pose a danger risk [5].

A possible solution that attempts to solve the limited capacity issue would be storing hydrogen gas into abandoned oil wells. Because old gas and oil wells are structured for natural gas and carbon dioxide storage, it is easier to transition abandoned oil wells into long term hydrogen gas storage underground. Using these oil wells as hydrogen storage is a viable option because it can store very large amounts of hydrogen gas while being able to reuse infrastructure [6]. Safety management for gas leaks through the capstones have also been heavily researched through its need for natural gas reserves. Abandoned oil wells are limited to their locations. Another solution for large underground hydrogen storage is salt caverns [7]. By modifying naturally occurring salt caverns, hydrogen gas can be stored like abandoned oil wells. Salt caverns are a good option as well because they are also geologically stable for storing hydrogen gas, but they are also limited to location. Instead of using hydrogen gas as storage, another possible method is storing it in liquid form. Cryogenic hydrogen storage takes hydrogen gas and lowers the temperature to -253˚C [8]. This process liquifies the hydrogen to store it. Liquid hydrogen is more stable than its gaseous form, and it is also energy dense. Because of this, cryogenic storage is preferred for transporting hydrogen and large hydrogen storage.

C. Synthetic Fuels and Green Ammonia

Synthetic fuels and green ammonia are another use for hydrogen [9]. These processes include storing hydrogen in synthetic fuels by reacting with carbon dioxide, or as green ammonia by reacting with nitrogen gas. Both compounds are much more stable compared to hydrogen gas alone, as they can liquefy at higher temperatures. However, both processes require significant energy to react and additional infrastructure in production, storage, and transportation to implement into the process.

D. Design Choice

Taking all technological options into account, this project works best with PEM electrolyzers and fuel cells and pressurized tanks for hydrogen storage. With a distributed energy management system, PEM electrolyzers and fuel cells will work best because of its quick ramp up times which are able to handle large load variability. PEM electrolyzers also output high purity hydrogen gas, which increases efficiency in the system when using hydrogen gas to create electricity. While alkaline electrolyzers have longer lifetimes and cheaper costs, they would be less efficient overall compared to PEM electrolyzers, and have slower start up times. For hydrogen storage, pressurized tanks would be the most versatile option. Some areas may not have the necessary locations for salt caverns or underground hydrogen storage, and cryogenic hydrogen storage costs more in infrastructure and energy. To an extent, pressurized tanks would be able to scale up for larger projects.

III. ALGORITHM

To demonstrate the impact of the hydrogen storage solution, an algorithm was developed to process real energy market data, simulate hydrogen generation and storage from excess solar energy, and record the outcomes for analysis. The objective is twofold. First, to see a reduction in the cost of importing non-renewable energy when solar generation is unable to meet demand. Second, that the generation of hydrogen from excess solar energy can produce revenue for users of the technology. The algorithm works by operating on three metrics collected for a single area over the course of a year: solar generation, demand, and Locational Marginal Pricing (LMP) data. Solar generation, measured in MWh, can be defined as the sum of electricity generated from photovoltaic and solar thermal-electric systems located within the utility's region of control. Demand is defined as the energy needs within the area. LMP represents the wholesale market price of electricity bought to sustain a grid.

LMP data was obtained from the Southwest Power Pool (SPP) via the Western Energy Imbalance Service (WEIS) Market [10]. Demand and solar generation data were collected from the U.S. Energy Information Administration (EIA) API [11]. Data from both sources were gathered at two-hour intervals, which themselves average values reported every fifteen minutes. The program collects data by making API requests to two sources: it downloads LMP data from the SPP portal using HTTP GET requests for specific file paths. It also retrieves solar generation, net generation, and demand data from the EIA API using structured queries with an API key. After fetching the data, it merges the LMP values with the corresponding EIA data for each day, consolidating them into a single table, which is then saved as a CSV file. To process the CSV file, the data is sorted in reverse-chronological order and then operated on sequentially.

A. Case Studies

El Paso Electric (EPE) was selected for analysis due to its balanced profile of strong solar generation at 2.39 million MWh per year and moderate energy demand of 5.6 million MWh, making it an ideal case for evaluating the benefits of hydrogen storage. In contrast, the Imperial Irrigation District (IID) was chosen for its massive solar output of 14.25 million MWh per year and low demand of 2.45 million MWh, offering insight into hydrogen storage potential for larger solar farms. In 2022, the California Independent System Operator (CAISO) curtailed 2.4 million MWh of utility-scale wind and solar energy, resulting in an estimated $800 billion loss [12]. To address these losses, Hydrolyte Technologies offers hydrogen storage solutions that allow energy producers to store excess electricity and reduce curtailment. This technology also extends to residential solar user, among the 4.2 million U.S. homes equipped with solar panels, who often generate surplus energy during low-demand periods [13].

The solar generation and demand data between March 2024 and March 2025 for the El Paso residential area can be seen in Figure 2, which shows solar generation (orange curve) exceeding demand (blue curve), highlighting the need for energy storage.

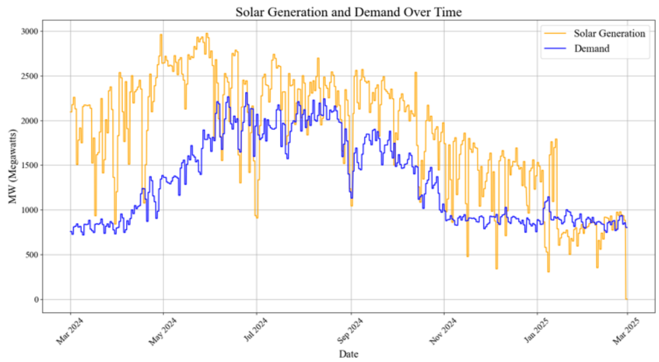


FIGURE 2: Solar generation and demand in El Paso over one year.

B. Algorithm Processing

Figure 3 documents the basic operations of the algorithm as well as the cost reduction/revenue calculations in flowchart form.

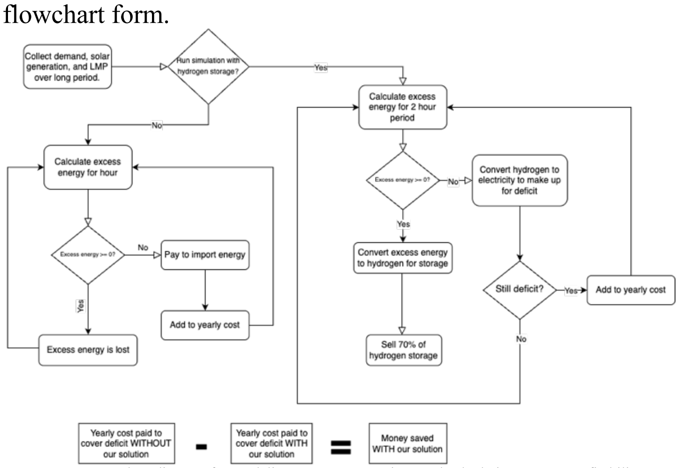


FIGURE 3: Flow diagram for modeling energy conversions and calculating system profitability.

The system's purpose is to convert excess solar energy to hydrogen, store this hydrogen, and supply it during high-demand, low-production periods, such as in July 2024. The algorithm calculates theoretical excess and deficit energy based on the collected solar generation and demand data. When there is an excess of energy (solar generation exceeding demand), it is converted into hydrogen and then "stored" at a rate of 1 MWh of energy per 17.54 kg of hydrogen. Hydrogen storage capacity is predefined as ten 1500kg tanks for a total capacity of 15,000kg. This total tank capacity was chosen to ensure that storage limitations would not be a constraint on the performance of the hydrogen-based DERMS. When there is a deficit of energy (demand exceeds solar generation), the algorithm "pulls" the required amount of hydrogen from storage and converts it back into electricity with a conversion rate of 59.92 kg of hydrogen per MWh. This rate is calculated using the lower heating value of hydrogen (33.3 kWh/kg), adjusted for the 50% fuel cell efficiency, converted from kWh to MWh, and inverted to express the required hydrogen mass per MWh of electricity produced.

Cost savings are achieved by stabilizing the grid during periods of low solar generation, thereby reducing the need to purchase non-renewable energy to cover energy deficits. This process is simulated using deficit energy calculations and Locational Marginal Pricing (LMP) data. The algorithm uses these inputs to estimate the cost of meeting energy demand through non-renewable energy imports. The total annual cost reduction is calculated as the difference between the cost of importing non-renewable energy with and without the DERMS in operation.

Revenue is calculated by "selling" hydrogen under favorable market conditions, where the system aims to sell up to 70% of stored hydrogen to external regions at a predetermined sale price, assuming there is a consistent demand for hydrogen. A minimum of 30% of the stored hydrogen is perpetually maintained to service the local area during periods of high demand. This way, the system maintains a reserve of hydrogen for potential shortages while capitalizing on excess supply throughout the year. The algorithm continuously monitors financial and energy-related metrics, generating visualizations to evaluate the long-term impact of hydrogen storage on overall system performance and profitability.

C. Data Analysis

After processing a year's worth of data, it is evident that the system significantly reduces costs and generates revenue for the producer. Figure 4 shows the relationship between solar generation, energy demand, and the financial impact of implementing this process over time. The background lines in orange and blue represent solar generation and energy demand, respectively. Periods where the orange curve (solar generation) falls below the blue curve (demand) indicate energy deficits—times when the grid must rely on alternative energy sources. In these deficit periods, the system activates by supplying stored hydrogen-generated electricity, effectively reducing the need to purchase non-renewable energy from the external market.

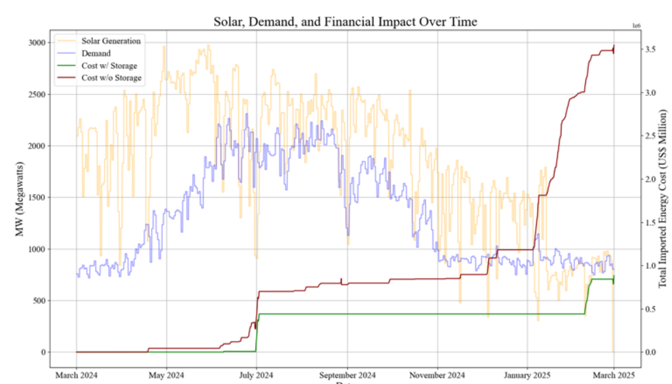


FIGURE 4: Financial impact of utilizing hydrogen storage system in EPE over one year.

This benefit is illustrated by the two cost curves. The green line represents the cumulative cost of energy imports with the system in place. The dark red line shows the cost without the system, where all deficits are covered by buying non-renewable energy. The sharp upward trend of the red curve during months like July 2024 and early 2025 highlights how quickly costs rise when solar generation cannot meet demand and no hydrogen storage is available. In contrast, the green curve rises much more gradually, reflecting energy supply stability and financial loss mitigation. Overall, the graph demonstrates that the system not only buffers the grid during periods of low solar production but also results in substantial cost savings by reducing dependency on expensive non-renewable energy imports.

Energy Savings and Revenues	**Electric Power Providers**	
	Imperial Irrigation District (Solar Farm)	**El Paso (Residential)**
Energy savings (/year)	14.25 million MWh	2.39 million MWh
Money saved from not outsourcing (/year)	$113,000	$2.6 million
Hydrogen sales revenue (/year)	$133 million	$22 million

TABLE 2: Energy savings and revenue earned from hydrogen storage system.

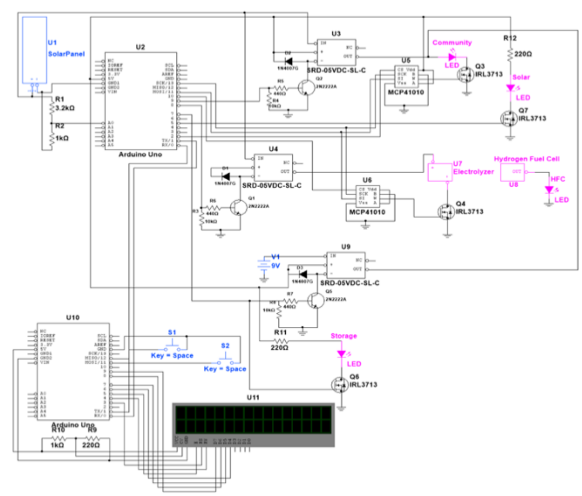
Although adding more tanks increases profits for both utilities, the optimal number of tanks varies due to their differing energy profiles. EPE, serving a densely populated residential area with limited excess energy, benefits the most with 10 tanks, as seen in Table 2, where cost savings and revenue increase substantially by saving $83,000 per year and making $22 million in revenue, while avoiding diminishing returns beyond this point. With a daily average energy excess of 562 MWh and efficiencies factored in, 10 storage tanks can store all excess energy throughout the day, making it an ideal storage size. IID, with its abundance of surplus energy from solar farms, can maximize profitability with 40 tanks, where the balance between cost savings and revenue remains favorable at $41,000 savings and $133 million increase in revenue. By utilizing hydrogen storage in energy shortages, the utility relies less on outsourcing non-renewable energies. In the one-year period, EPE saved a total of $2.6 million while IID saved $113,000. The difference in savings is related to the contrasting energy demands within a large residential community like El Paso, compared to a less populated area like the Imperial County. Beyond these tank sizes, additional tanks yield diminishing returns, making 10 tanks ideal for EPE and 40 tanks optimal for IID to ensure efficient hydrogen storage and profitability.

IV. BENCH SCALE

A. System Introduction

A bench-scale system was developed to both validate and demonstrate the functionality and control of the grid-firming solution at a small low-risk scale. By simulating four different scenarios the system responds appropriately depending on its dynamic inputs which effectively shows how the design can be scaled. The objective was to maintain an uninterrupted and consistent flow of power to a designated source. In this system, it has been portrayed by the Community LED that maintains its brightness. The bench-scale system is illustrated in Figure 5.

A)



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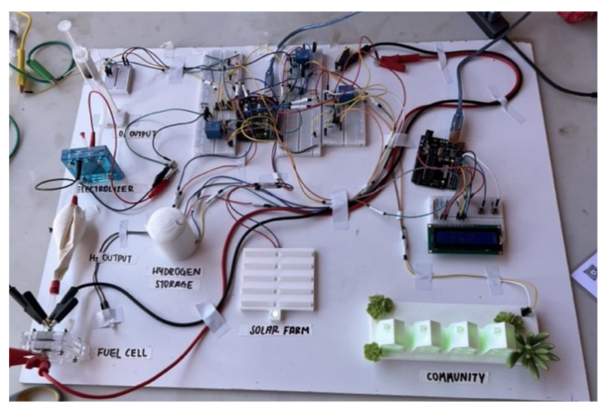

FIGURE 5: Bench-scale implementation shown as (A) Technical Connections and (B) Actual Set-up (solar panel is not pictured).

Blue peripherals and pink peripherals display the systems inputs and outputs respectively. Inputs consist of a small-scale photovoltaic that produces at peak hours ~20V, two tactile push buttons for user-defined demand values, and a 9V battery. The outputs include a series of LEDS including the Community LED, a 16x2 LCD screen, an electrolyzer, and a hydrogen fuel cell. The system includes two Arduino Uno microcontrollers, MCP41010 digital potentiometers, SRD-05VDC-SL-C relay switches, 1N4007 diodes, 2N2222 NPN BJTs, and STP40NF03L N-Channel MOSFETs. A variety of different resistors, breadboards, and wires were used to make a complete system. Due to efficiency, budget, and safety constraints the system does not include a storage system, instead it uses a 9V battery in the event of needing to "pull from storage.

Arduino Uno microcontrollers performed real time monitoring, control, and user interface functions. One microcontroller continuously sampled the solar panels voltage via an on-board analog-to-digital converter (ADC). This microcontroller processed the input signal, actuated relays, adjusted potentiometers, and regulated power flow. The other microcontroller supported the user-friendly interface, displaying both supply and demand values onto an LCD screen. Two tactile buttons provided the user with complete control over the load's theoretical demand. Microcontrollers exchanged their sampled input values with each other over direct UART communication at a baud rate of 9600bps.

Power was limited and directed throughout the system correctly using digital potentiometers, relay switches, and MOSFETs. Relay switches functioned as electronically controlled gates to either enable or cutoff current flow. These isolated different regions of the system depending dynamically on specific conditions. The solar panel output, which reached approximately 20V, required proper regulation to protect sensitive components of the system. To achieve an electronically controlled variable, a digital potentiometer controlled the gate voltage of a MOSFET series connected to the load.

The digital potentiometers communicate via SPI with the microcontroller, allowing the algorithm to control the wiper position. Each digital potentiometer was set up into a voltage divider configuration with a 5V supply. The voltage at the wiper was connected to the gate of a corresponding MOSFET. Powering the gate alters the gate-to-source voltage (V_{GS}) which was controlled and used to place the MOSFET into the triode region. Within this region the MOSFET can be manipulated to act as a variable resistor. Increasing the V_{GS} increases drain current (I_D), resulting in a decreased resistance value. While decreasing the gate voltage increases resistance by limiting I_D.

B. System's Operation

The system has four different cases and is clearly identifiable via the LEDs. The LEDs display the source(s) that are being used to support the community demand. Explaining the switches within each case will be based on their numbering found in figure 5A and will refer to them as either open or closed. Open switches have infinite resistance and act as an open circuit, while closed switches enable current flow, also known as off and on respectively.

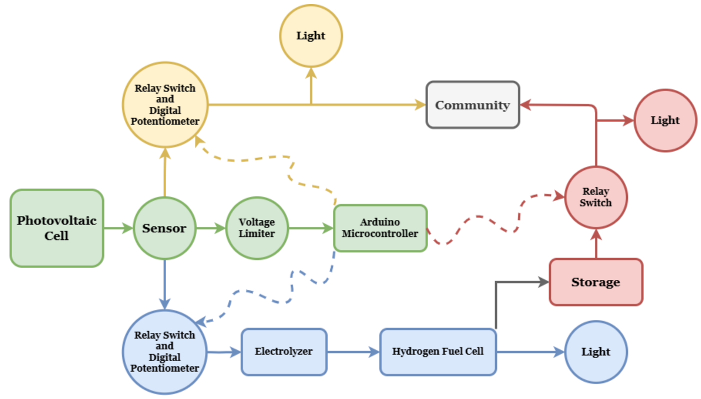

FIGURE 6: Flow diagram of bench-scale implementation.

As mentioned above the system can be split into four different operational cases which will be explained with Fig 6. The first case being that represents insufficient solar power, relays U3 and U4 open, preventing solar power flow. Relay U9 is closed, enabling the Community LED to be powered by the battery, indicated by the Storage LED. The green and red sections in Fig. 6 depict this case. In Case 2, where the solar panel provides partial power, relays U3 and U4 are still open and U9 is closed. Allowing the Community LED to receive power from just the battery. However, the storage LED and solar LED are both active in this case to indicate battery usage and solar contribution. The system provides this abstraction to display the right functionality without actually having a hardware voltage summer. As shown in Fig 6, this corresponds to the green, red, and yellow sections.

Case 3 signifies optimal solar power, where only relay U3 is closed, fully supplying the Community LED directly from the solar panel. In this scenario, the Solar LED is on, indicating exclusive solar power usage. Displayed in Fig. 6 as green and yellow. In Case 4, excess solar power conditions prevail and relays U3 and U4 are closed, while relay U9 is open, enabling surplus energy diversion. The colored sections blue, green and yellow in Fig. 6 demonstrate this case. The Community LED remains powered directly from the solar panel, and the Solar LED is active. "Excess" energy is redirected to an electrolyzer, initiating electrolysis in deionized water, producing hydrogen and oxygen gases. Oxygen is released into the atmosphere, while hydrogen is fed into a hydrogen fuel cell (HFC). The fuel cell recombines hydrogen with atmospheric oxygen, generating water and electricity, which activates the HFC LED.

C. Design Decisions and Conclusion

In Case 2 it was mentioned that even though solar power was available it wasn't used to contribute to the load. This was an intentional design decision aimed at conceptually demonstrating two source summation, without having the hardware to do so. This preserved the main idea for the case while avoiding any electrical conflicts between unsummed voltages. Case 4 also mentions redirecting "excess" energy which in an ideal situation would be correct, however, due to scale this is just not possible. The electrolyzer operates at 1.8V and is limited in speed. Any excess voltage higher than 1.8V would need to be burned off, the system is rather an abstracted version of a large scale. Case 4 is primarily a demonstrative idea that both the community and the electrolyzer can be used due to excess. The hydrogen fuel cell is not supposed to be a part of this case but due to storage limitations, the fuel cell is used to show that the idea works. Also, demand is just a value used to compare with solar production to decide the case. Our load is just an LED, which doesn't require lots of power, therefore this is also purely demonstrative.

The bench-scale implementation proved to be a small-level demonstration of a power firming system. Although not a full DERMS system, it provides a proof-of-concept simulation of what a DERMS should do. By maintaining constant brightness to the "Community LED" regardless of any input fluctuation, the system delivered reliable and uninterrupted power. The system's behavior reflects how, in a full-scale system, stored hydrogen and a fuel cell could generate enough energy to support the demands of the grid. Similarly, during times of excess, an electrolyzer and pressurized storage could be used to store energy. The system validated key concepts of power firming and system switching for successful scaling to a grid-level application.

V. DISCUSSION

Hydrogen-based DERMS are proving themselves to have potential in supporting the shift to clean energy on the large scale. In its present state, PEM electrolyzer and fuel cell technology is fully capable of being implemented for use in energy storage and grid support. Pressurized storage tanks are both reliable and scalable, making them useful for long-term storage to supplementing the grid. These devices allow for seamless conversions between electricity and hydrogen through clean processes, minimizing environmental impact. This system may play a key role in the pivot towards clean energy.

The energy storage market is growing quickly, with new installations in the U.S. in 2024 combining for over 37 GWh of storage, a 34% increase over 2023 [14]. This trend is likely to continue, as increasing sustainable energy creates greater variability in energy production, elevating the need for energy storage options to avoid curtailment and losses. As the demand for energy storage rises, more resources will be put towards improving the efficiencies and capital costs of key hydrogen components such as electrolysis systems, fuel cells, and storage methods. These costs will become more competitive with battery storage facilities, with the additional benefits of being environmentally friendly and enabling longer-term storage in higher quantities.

Achieving a successful carbon-neutral grid is heavily reliant on finding new ways to store and distribute renewable energy. Utilization of electrolyzers, compressed-gas storage tanks, and fuel cells is a feasible way to improve reliability and efficiency of solar energy. These devices are not new concepts, but continuing research and development of these technologies is important for lowering costs and increasing widespread access to clean energy.

VI. CONCLUSION

Integrating hydrogen-based technologies with battery storage presents a highly flexible and technically robust solution to the critical challenge of renewable energy intermittency. Hydrogen systems, through water electrolysis and fuel cell regeneration, enable long-duration energy storage that complements the fast-response, high-efficiency performance of lithium-ion batteries. This hybrid approach is particularly suited for large-scale deployments like the Imperial Irrigation District, where surplus renewable energy can be converted into hydrogen and stored for weeks or even months, ensuring firm power delivery during low-generation periods. Similarly, residential-scale applications, such as those analyzed in El Paso, benefit from hydrogen's ability to balance grid fluctuations over longer durations than battery systems alone can manage. Hydrogen's high energy density by mass and the modularity of pressurized storage infrastructure make it a strategic asset in scaling renewable energy systems to meet diverse demand profiles while improving overall grid stability.

The bench-scale implementation served as a critical proof-of-concept for these principles. Although not a complete Distributed Energy Resource Management System (DERMS), the project effectively simulated key functions of such a system. The hydrogen-powered setup demonstrated its ability to maintain consistent power delivery—such as keeping the "Community LED" at constant brightness—despite fluctuations in energy input. This small-scale success reflects how, at the utility level, hydrogen storage and fuel cells can step in to meet grid demand during periods of low renewable generation, while electrolyzers and pressurized storage tanks capture and store excess energy during peak production times. The system validated essential aspects of power firming, seamless switching between energy sources, and the practical feasibility of scaling hydrogen storage technologies to grid-level operations.

As global energy markets rapidly evolve and more countries and corporations commit to transitioning toward 100% renewable energy fleets, the role of hydrogen and other innovative storage systems becomes increasingly vital. With solar and wind generation expanding at unprecedented rates, pairing these sources with reliable, long-duration storage, such as green hydrogen, will be key to ensuring energy security and grid resilience. The future of clean energy depends not only on generating renewable power but also on building robust systems that can store and dispatch it effectively. Hydrogen technologies, as demonstrated in this project, are poised to play a central role in realizing that vision.

VII. ACKNOWLEDGMENT

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